

Response to P4043R0

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R0: March 2026 (Croydon)

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Abstract

P4043R0^[1] ("Are C++ Contracts Ready to Ship in C++26?") proposes that EWG reconsider the shipping vehicle for C++ Contracts. Every paper it cites, every concern it raises, and every alternative it suggests was available to the committee before the most recent decisions on the topic.

Revision History

R0: March 2026 (Croydon)

- Initial version.
 - Corrected imprecise ballot characterization in Section 4 (“none has achieved consensus” replaced with “each was rejected”) and added Section 5 on the distinction between consensus against and no consensus.
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1. Disclosure

The author has no involvement in the C++ Contracts facility and holds no position on its design. The question addressed here is narrow: does [P4043R0](#)^[1] present information that the committee has not already considered?

2. The Question

[P4043R0](#)^[1] asks whether EWG should reconsider the shipping vehicle for C++ Contracts. The paper asks this question seven times:

Section	Representative statement
1	“This paper therefore proposes a simple question for EWG: whether the C++ Contracts facility, in its current form, is ready to ship in C++26.”
2	“The presence of continuing NB discussion suggests that further design exploration may still be beneficial.”
3	“Taken together, this body of work indicates that the design space for Contracts continues to be actively discussed.”
4	“If substantial questions remain open, deferring the feature could allow the committee to incorporate additional design insights.”
5	“If the committee concludes that the C++ Contracts facility would benefit from additional design time, several alternative paths could be considered.”
6	“Do we believe that the C++ Contracts facility is ready to ship in C++26?”
7	“Deferring the feature could allow WG21 additional time to refine the design.”

3. The Evidence

P4043R0^[1] cites 15 papers. The following table lists each one with its publication date, drawn from the paper's own reference section:

Paper	Published
P2900R14	2025-02-13
P3573R0	2025-01-12
P3829R0	2025-07-15
P3835R0	2025-09-03
P3851R0	2025-09-29
P3853R0	2025-10-05
P3909R0	2025-11-02
P3911R2	2026-01-14
P3912R0	2025-12-15
P3919R0	2025-11-05
P3946R0	2025-12-14
P4005R0	2026-02-02
P4009R0	2026-02-09
P4015R0	2026-02-10
P4020R0	2026-02-23

Every paper in this list was published before P4043R0^[1] and was available at prior committee meetings. The paper does not cite any new implementation result, deployment observation, or design analysis.

4. The Record

On 2025-02-11, EWG polled on whether to remove P2900^[2] (“Contracts for C++”) from consideration for C++26 and find a different shipping vehicle. The result was consensus against (P2899R1^[3]).

On 2025-02-13, LWG forwarded P2900R14^[2] to plenary with consensus (P2899R1^[3]).

On 2025-02-16, plenary adopted P2900R14^[2] into the C++26 Working Paper with strong consensus (P2899R1^[3]).

P2900R14^[2] (published 2025-02-13) remains the latest revision. The changes applied since adoption - the removal of `evaluation_exception()` via P3819R0^[9] and the addition of a feature test macro via P3886R0^[10] - are NB comment resolutions that narrow or clarify the facility. The core design is unchanged. Multiple proposals to modify the design have been presented since adoption; each was rejected (P3846R0^[4]).

`constexpr` shipped as a minimal language feature in C++11 (N2235^[5]). C++14 relaxed its constraints dramatically (N3652^[6]). The MVP model - ship a minimal language feature in one standard, expand it in the next - is established WG21 practice.

5. Framing

A committee poll produces one of three outcomes: consensus for, consensus against, or no consensus. These are not interchangeable. Consensus against a proposal is an affirmative decision - the committee considered the question and answered it. No consensus means the committee did not reach a decision in either direction.

P4043R0^[1] describes prior ballot outcomes without distinguishing between these cases. The EWG poll on removing Contracts from C++26 was not “no consensus.” It was consensus against removal. The proposals to modify the adopted design were not merely unsuccessful - they were rejected. Characterizing consensus against as the absence of consensus erases the committee’s decision.

6. The Paradox

P4043R0^[1] recommends deferral to allow “implementation and deployment experience.” Language features do not receive widespread deployment outside standards. The Concepts TS (ISO/IEC TS 19217:2015^[7]) demonstrated this: the TS was withdrawn, and Concepts were redesigned and shipped directly in C++20.

P4043R0^[1] prescribes the condition it claims to remedy.

7. Procedure

P4043R0^[1] (dated 2026-03-07) does not appear in any official WG21 mailing. The 2026-02 pre-Croydon mailing (released 2026-02-23) contains papers through **P4032R0**^[8]. The paper presents no proposed solution to the one outstanding NB comment (RO 2-056).

References

1. **P4043R0** - "Are C++ Contracts Ready to Ship in C++26?" (Dariusz Neațu, 2026). <https://wg21.link/p4043r0>
2. **P2900R14** - "Contracts for C++" (Joshua Berne, Timur Doumler, Andrzej Krzemiński, et al., 2025). <https://wg21.link/p2900r14>
3. **P2899R1** - "Contracts for C++ - Rationale" (Timur Doumler, Joshua Berne, Andrzej Krzemiński, Rostislav Khlebnikov, 2025). <https://wg21.link/p2899r1>
4. **P3846R0** - "C++26 Contracts, reasserted" (Timur Doumler, Joshua Berne, 2025). <https://wg21.link/p3846r0>
5. **N2235** - "Generalized Constant Expressions" (Gabriel Dos Reis, Bjarne Stroustrup, Jens Maurer, 2007). <https://open-std.org/jtc1/sc22/wg21/docs/papers/2007/n2235.pdf>
6. **N3652** - "Relaxing constraints on constexpr functions" (Richard Smith, 2013). <https://open-std.org/jtc1/sc22/wg21/docs/papers/2013/n3652.html>
7. ISO/IEC TS 19217:2015 - "C++ Extensions for Concepts" (ISO, 2015). Withdrawn.
8. **P4032R0** - last paper in the 2026-02 pre-Croydon mailing.
9. **P3819R0** - "Remove evaluation_exception from Contracts" (Timur Doumler, Joshua Berne, 2025). <https://wg21.link/p3819r0>
10. **P3886R0** - "Contract violation handler replaceability" (Timur Doumler, Joshua Berne, 2025). <https://wg21.link/p3886r0>
11. **P3573R0** - "Contract concerns" (Michael Hava, J. Daniel Garcia Sanchez, et al., 2025). <https://wg21.link/p3573r0>
12. **P3829R0** - "Contracts do not belong in the language" (David Chisnall, John Spicer, et al., 2025). <https://wg21.link/p3829r0>
13. **P3835R0** - "Contracts make C++ less safe - full stop!" (John Spicer, Ville Voutilainen, Jose Daniel Garcia Sanchez, 2025). <https://wg21.link/p3835r0>
14. **P3851R0** - "Position on contract assertions for C++26" (J. Daniel Garcia, et al., 2025). <https://wg21.link/p3851r0>
15. **P3853R0** - "A thesis+antithesis=synthesis rumination on Contracts" (Ville Voutilainen, 2025). <https://wg21.link/p3853r0>

16. **P3909R0** - "Contracts should go into a White Paper" (Ville Voutilainen, 2025). <https://wg21.link/p3909r0>
17. **P3911R2** - "Make Contracts Reliably Non-Ignorable" (Darius Neațu, Andrei Alexandrescu, et al., 2026).
<https://wg21.link/p3911r2>
18. **P3912R0** - "Design considerations for always-enforced contract assertions" (Timur Doumler, et al., 2025).
<https://wg21.link/p3912r0>
19. **P3919R0** - "Guaranteed-(quick-)enforced contracts" (Ville Voutilainen, 2025). <https://wg21.link/p3919r0>
20. **P3946R0** - "Designing enforced assertions" (Andrzej Krzemieński, 2025). <https://wg21.link/p3946r0>
21. **P4005R0** - "A proposal for guaranteed-(quick-)enforced contracts" (Ville Voutilainen, 2026).
<https://wg21.link/p4005r0>
22. **P4009R0** - "A proposal for solving all of the contracts concerns" (Ville Voutilainen, 2026).
<https://wg21.link/p4009r0>
23. **P4015R0** - "Enforcing Contract Conditions with Statements" (Lisa Lippincott, 2026).
<https://wg21.link/p4015r0>
24. **P4020R0** - "Concerns about contract assertions" (Andrzej Krzemieński, 2026). <https://wg21.link/p4020r0>